Try these

1. What percentage of a day is 10 hours?
2. Divide ₹350 among P, Q and R such that P gets 50% of what Q gets and Q gets 50% of what R gets.

Think

With a lot of pride, the traffic police commissioner of a city reported that the accidents had decreased by 200% in one year. He came up with this number by stating that the increase in accidents from 200 to 600 is clearly a 200% rise and now that it had gone down from 600 last year to 200 this year should be a 200% fall. Is this decrease from 600 to 200, the same 200% as reported by him? Justify.



Example 4.5

The income of a person is increased by 10% and then decreased by 10%. Find the change in his income.

Solution:

Let his initial income be ₹ x.

Income after 10% increase is

$$₹\,x + \left(\frac{10}{100} \times x\right) = ₹\,\frac{110}{100}\,x \text{ (or) } ₹\,\frac{11}{10}\,x$$

Now, income after 10% decrease is,

$$₹\,\frac{11x}{10} - \frac{10}{100}\left(\frac{11x}{10}\right)$$

$$\frac{11x}{10} - \frac{11x}{100} = \frac{110x - 11x}{100} = ₹\,\frac{99x}{100}$$

$\therefore$ Net change in his income $= x - \frac{99x}{100} = \frac{x}{100}$

$\therefore$ Percentage change $= \frac{\frac{x}{100}}{x} \times 100\% = 1\%$

That is, income of the person is reduced by 1%.

(or)

Aliter:

Let his income be ₹100

Income after 10% increase is

$$100 + 100 \times \frac{10}{100} = ₹110.$$

Now, income after 10% decrease is,

$$110 - 110 \times \frac{10}{100} = 110 - 11 = ₹99$$

$\therefore$ Net change in his income = 100 – 99 = 1

$\therefore$ Percentage change $= \frac{1}{100} \times 100\% = 1\%$.

That is, income of the person is reduced by 1%.

Note

For any given number, when it is increased first by x % and then decreased by y %, then the value of the number is increased or decreased by $\left(x + y + \frac{xy}{100}\right)\%$. Use 'negative' sign for decrease and also assume 'decrease' if the sign is negative. Use this note to check the answer for the Example 4.5.

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